

You are a senior hedge-fund quantitative trading architect, institutional signal-engine reviewer, TypeScript engineer, machine learning engineer, adaptive AI trading specialist, probabilistic systems designer, market microstructure analyst, and capital-risk management expert.

I already have an existing trading signal system.

The system currently generates market signals such as:
BUY / SELL / NO TRADE

It may support:

- stocks
- crypto
- spot trading
- futures trading

My objective is NOT to create a toy trading bot.

My objective is NOT to fake 95% accuracy.

My objective is NOT to hardcode confidence.

My objective is NOT to force BUY/SELL signals.

My real objective is to turn the existing system into a serious, signal-only, institutional-grade trading signal intelligence engine that focuses on:

- capital protection first
- fewer but higher-quality A+ setups
- low false-signal rate
- strong risk/reward
- realistic confidence scoring
- positive expectancy
- deep NO TRADE intelligence
- bullish/bearish directional intelligence
- multi-timeframe confirmation
- spot/futures separation
- futures liquidation safety
- market-regime awareness
- macro/correlation/sentiment safety filters
- realistic backtesting
- walk-forward validation
- out-of-sample validation
- paper signal journaling
- self-learning from failed signals without overfitting
- continuous improvement using statistically valid feedback

Critical system rule:

This system must remain SIGNAL-ONLY.

Do not create automatic live trade execution.

Do not place live orders.

Do not connect broker execution.

Do not create Binance/Bybit futures order placement.

Do not create Alpaca/IBKR order placement.

Do not create market-buy or market-sell execution.

The system can output trading signals and trade decisions, but the human trader will execute manually.

Very important:

A BUY signal is not always an executable trade.

A SELL signal is not always an executable trade.

Every final signal must be classified as:

- TAKE
- WAIT
- SKIP
- NO TRADE

The system must distinguish between:

- watchlist alert
- weak signal
- conditional setup
- executable A/A+ setup

Do not assume indicators mean profitability.

Do not assume many candle patterns mean profitability.

Do not assume many strategies mean profitability.

Do not assume historical optimization means future edge.

Do not overfit.

Do not blindly retrain after one failed signal.

Do not increase confidence without validation.

Prefer NO TRADE when conditions are unclear.

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MASTER WORKFLOW RULE

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Work in this exact order:

STAGE 1 — FORENSIC AUDIT ONLY

First inspect the entire existing codebase.

Do not implement.

Do not edit files.

Do not refactor.

Only discover, map, score, and report.

STAGE 2 — IMPLEMENTATION PLAN

After the audit, produce a phase-wise implementation plan based on actual gaps found.

Do not implement yet.

STAGE 3 — PHASE-WISE IMPLEMENTATION

Only after I explicitly say "START IMPLEMENTATION", implement missing features phase by phase.

Do not jump randomly.

Do not rewrite everything unless the architecture is truly unusable.

Preserve existing working code.

Add compatibility adapters if needed.

STAGE 4 — VALIDATION

After implementation, validate through:

- unit tests
- integration tests
- backtesting
- walk-forward testing
- out-of-sample testing
- paper signal journal

STAGE 5 — SELF-LEARNING

Only after backtesting and journaling exist, implement adaptive learning.

Do not build fake learning before outcome tracking exists.

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STAGE 1 — FORENSIC AUDIT
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PHASE 1 — COMPLETE CODEBASE DISCOVERY
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Scan the entire repository deeply.

Inspect all folders including:

- src
- app
- lib
- server
- services
- utils
- strategies
- indicators
- patterns
- engines
- models
- ml
- ai
- data
- providers
- config
- scripts
- tests
- docs
- README files
- API routes
- CLI commands
- UI components

Search aggressively for every hidden trading-related logic using keywords:

candle
candlestick
OHLC
OHLCV
open
high
low
close
volume
signal
buy
sell
hold
no trade
trade
strategy

indicator
confidence
probability
score
bullish
bearish
neutral
trend
momentum
reversal
breakout
support
resistance
liquidity
sweep
order block
fair value gap
FVG
Fibonacci
Ichimoku
Supertrend
EMA
SMA
WMA
HMA
RSI
MACD
VWAP
ATR
ADX
Bollinger
Stochastic
CCI
ROC
OBV
MFI
pivot
swing high
swing low
stop loss
take profit
trailing stop
risk
risk reward

backtest
paper trade
journal
outcome
feedback
train
retrain
model
learning
weights
calibration
Binance
Coinbase
Yahoo
Alpha Vantage
Polygon
Twelve Data
Alpaca
Bybit
leverage
margin
liquidation
funding

Create an exhaustive inventory table with:

- file path
- module/function/class name
- category
- what it does
- inputs
- outputs
- active or unused
- where it is imported/called
- whether connected to final signal engine
- complete or stubbed
- real or mocked
- hardcoded or dynamic
- tested or untested
- production risk level
- recommended action

Categories must include:

- market data provider

- candle validator
- candlestick pattern
- technical indicator
- price action module
- market structure module
- bullish/bearish bias engine
- market regime engine
- strategy module
- confluence engine
- confidence scoring
- probability scoring
- risk engine
- futures risk engine
- spot/futures mode handler
- final signal engine
- trade decision layer
- live signal grading
- basket expectancy
- portfolio/exposure risk
- macro/sentiment filter
- backtest engine
- validation engine
- signal journal
- outcome tracker
- self-learning system
- model training
- model versioning
- audit/replay logger
- API/CLI/UI output
- config
- tests

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PHASE 2 — SIGNAL-ONLY SAFETY AUDIT

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Verify that the system is signal-only.

Check for:

- automatic live order execution
- broker placement
- exchange order placement
- Binance futures order execution
- Bybit order execution

- Alpaca order execution
- IBKR order execution
- market buy/sell commands
- limit/stop order submission
- API keys used for trading
- hidden execution services
- auto-trading loops

If any live execution exists, mark HIGH RISK.

Required final principle:

The system may produce:

- SPOT_BUY
- SPOT_SELL_EXIT
- SPOT_NO_TRADE
- FUTURES_LONG
- FUTURES_SHORT
- FUTURES_NO_TRADE

But it must not execute orders automatically.

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PHASE 3 — BULLISH / BEARISH MARKET BIAS AUDIT

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Audit whether the system has true directional intelligence or only random indicator voting.

Check for these layers:

1. Candle Bias

For every detected candle pattern, verify whether it classifies:

- bullish reversal
- bearish reversal
- bullish continuation
- bearish continuation
- neutral indecision

2. Trend Bias

Check whether the system detects:

- bullish trend
- bearish trend
- sideways trend
- weak bullish

- weak bearish

Using:

- higher highs
- higher lows
- lower highs
- lower lows
- EMA stack
- SMA stack
- slope
- trendline reaction
- breakout/retest
- break of structure
- change of character

3. Momentum Bias

Check if buyer/seller pressure is measured using:

- RSI direction
- MACD histogram
- ADX strength
- volume expansion
- candle body strength
- momentum acceleration
- volatility expansion

4. Higher Timeframe Bias

Check if lower timeframe signals are validated against higher timeframe signals.

Required mapping:

- 1m checks 5m and 15m
- 5m checks 15m and 1h
- 15m checks 1h and 4h
- 1h checks 4h and 1d
- 4h checks 1d and 1w

5. Composite Directional Score

Check whether final engine calculates:

- bullishScore
- bearishScore
- neutralScore
- conflictScore

or equivalent.

Required behavior:

- BUY/LONG only if bullish score clearly dominates
- SELL/SHORT only if bearish score clearly dominates
- NO TRADE if mixed
- NO TRADE if conflict score is high
- NO TRADE if higher timeframe strongly disagrees
- NO TRADE if market regime is choppy

Flag HIGH RISK if:

- one weak indicator can trigger BUY/SELL alone
- bullish/bearish score is guessed
- confidence is arbitrary
- higher timeframe is ignored
- mixed signals still produce BUY/SELL

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PHASE 4 — CANDLESTICK PATTERN AUDIT

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Discover every implemented candle pattern.

Do not only search for common names.

Also detect custom candle logic such as:

- wick rejection
- long wick
- body strength
- buyer pressure
- seller pressure
- momentum candle
- exhaustion candle
- breakout candle
- range candle
- inside candle
- outside candle

Compare implementation against this full list:

Basic patterns:

- Doji
- Spinning Top
- Marubozu
- Hammer
- Inverted Hammer

- Shooting Star
- Hanging Man

Engulfing / bar structure:

- Bullish Engulfing
- Bearish Engulfing
- Inside Bar
- Outside Bar

Multi-candle patterns:

- Morning Star
- Evening Star
- Three White Soldiers
- Three Black Crows
- Tweezer Top
- Tweezer Bottom
- Piercing Pattern
- Dark Cloud Cover
- Bullish Harami
- Bearish Harami

Price-action candles:

- Pin Bar
- Rejection Candle
- Breakout Candle
- Exhaustion Candle
- Momentum Candle
- Long Wick Candle
- Narrow Range Candle
- Wide Range Candle
- Trap Candle
- Failed Breakout Candle

For each pattern, report:

- present or missing
- file/function
- formula used
- formula quality
- bullish/bearish/neutral classification
- reversal/continuation/indecision classification
- strength score exists?
- reliability score exists?
- explanation exists?
- timestamp/candle index exists?

- active in final engine?
- tested?
- profitability usefulness

Important:

Pattern name alone is not enough.

The system must understand context.

Example:

Bullish engulfing near support with volume and HTF alignment is meaningful.

Bullish engulfing inside strong bearish trend with no volume can be a trap.

Flag HIGH RISK if candle patterns are used without:

- support/resistance context
- trend context
- volume context
- higher timeframe context

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PHASE 5 — TECHNICAL INDICATOR AUDIT

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Discover all implemented indicators.

Compare against:

Trend indicators:

- SMA
- EMA
- WMA
- HMA
- Supertrend
- Ichimoku
- Parabolic SAR

Momentum indicators:

- RSI
- MACD
- Stochastic RSI
- Stochastic Oscillator
- CCI
- ROC
- Momentum

Volatility indicators:

- ATR
- Bollinger Bands
- Keltner Channels
- Donchian Channels
- Standard Deviation
- Bollinger Band Width

Volume indicators:

- VWAP
- OBV
- Volume Moving Average
- Volume Spike
- MFI
- Accumulation/Distribution

Key level indicators:

- Pivot Points
- Fibonacci Retracement
- Fibonacci Extension
- Swing High/Low
- Previous High/Low
- Previous Day High/Low
- Supply/Demand Zones

For each indicator, report:

- present or missing
- file/function
- formula correctness
- configurable period or hardcoded
- output value
- interpretation logic
- bullish/bearish/neutral output
- strength score
- active in strategy?
- active in final engine?
- tested?
- risk of false signal

Flag HIGH RISK if:

- indicator formula is wrong
- values are hardcoded
- interpretation is arbitrary

- one indicator alone can trigger a trade
- no tests exist for formula correctness

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PHASE 6 — STRATEGY INTELLIGENCE AUDIT

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Discover every strategy and hidden decision rule.

Search for:

- if/else trading rules
- score rules
- confidence formulas
- threshold-based decisions
- strategy config
- API-only logic
- UI-only strategy names
- backtest-only strategies

Compare against:

Core strategies:

- EMA crossover
- SMA crossover
- RSI reversal
- MACD momentum
- Bollinger mean reversion
- VWAP reclaim/rejection
- Support/resistance breakout
- Pullback continuation
- Candlestick reversal
- Trend following
- Volume breakout
- ATR breakout
- Multi-timeframe alignment

Advanced strategies:

- Liquidity sweep reversal
- Order block reaction
- Fair value gap reaction
- Fibonacci retracement bounce
- Opening range breakout
- Gap fill
- Break and retest

- Range breakout
- Range fade
- Scalping
- Swing continuation
- Momentum continuation
- Volatility contraction breakout
- Failed breakout reversal
- Retest continuation

For every strategy, report:

- name
- file path
- active or unused
- rules used
- indicators used
- candle patterns used
- market structure used
- required confirmations
- passed confirmations
- failed confirmations
- confidence logic
- stop loss logic
- take profit logic
- invalidation logic
- risk/reward logic
- market regime filter
- higher timeframe filter
- backtestable?
- historical performance available?
- tested?
- profitability trust score

Important:

Strategies must produce candidates only.

The final confluence engine decides whether the trade is executable.

Flag HIGH RISK if:

- strategy directly forces BUY/SELL
- strategy ignores market regime
- strategy ignores risk/reward
- strategy has no invalidation
- strategy is not backtested

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PHASE 7 — MARKET REGIME AUDIT

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Audit whether the system detects market condition.

Required regimes:

- bullish trending
- bearish trending
- sideways/ranging
- choppy/noise
- breakout regime
- mean reversion regime
- high volatility panic
- low volatility dead market
- low liquidity
- news shock regime

Check whether regime is calculated using:

- ADX
- ATR
- Bollinger Band Width
- volume
- trend slope
- range compression
- breakout expansion
- candle structure
- volatility expansion/contraction

Required behavior:

- trend strategies active in trending market
- mean reversion active in range
- breakout strategy active only with volume expansion
- choppy/noisy regime blocks or reduces signals
- high volatility panic forces stricter risk control
- low liquidity reduces confidence

Flag HIGH RISK if:

- same strategies run in all regimes blindly
- system does not identify chop
- system trades mean reversion during breakout
- system trades breakout during dead volume

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PHASE 8 — FINAL SIGNAL ENGINE AUDIT

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Find the exact final signal-generation file/function.

Explain:

- how BUY is generated
- how SELL is generated
- how NO TRADE is generated
- how SPOT signals are generated
- how FUTURES signals are generated
- all modules feeding final decision
- weighting formula
- confidence formula
- probability formula
- confluence formula
- no-trade gates
- conflict handling
- bad data rejection
- low risk/reward rejection
- low volume rejection
- high volatility rejection
- higher timeframe conflict rejection
- late entry rejection
- unvalidated strategy confidence cap

Classify the engine as:

- toy indicator bot
- basic signal aggregator
- intermediate strategy engine
- advanced selective signal engine
- institutional-grade probability engine

Flag HIGH RISK if:

- final signal is simple indicator voting
- confidence is guessed
- no NO TRADE gate exists
- weak signals become executable trades
- futures signals use same rules as spot
- no stop loss/invalidation required

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PHASE 9 — LIVE SIGNAL EXECUTABILITY AUDIT

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Audit actual generated signals as real trades.

For every generated signal, calculate:

- symbol
- asset type
- trading mode
- signal type
- entry price
- current price
- stop loss
- TP1
- TP2
- unrealized gain/loss %
- potential gain to TP1 %
- potential gain to TP2 %
- potential loss to stop %
- risk/reward ratio
- entry-to-current distance
- current price distance to target
- current price distance to stop
- whether entry is late
- whether setup is overextended
- whether setup is still valid
- whether setup should be skipped

For every signal, output:

- setup quality: A+ / A / B / C / Avoid
- trade decision: TAKE / WAIT / SKIP / NO TRADE
- reason

Setup grading:

A+ Setup:

- strong bullish/bearish confluence
- higher timeframe aligned
- strong volume confirmation
- risk/reward above 2
- clean invalidation
- not overextended
- market regime supports strategy
- historically validated

A Setup:

- good confluence
- acceptable volume
- risk/reward above 1.8
- minor conflict only

B Setup:

- tradable but not ideal
- some missing confirmation
- moderate confidence
- requires manual review

C Setup:

- weak signal
- low confidence
- unclear trend
- poor volume
- usually skip

Avoid:

- low confidence
- bad data
- poor risk/reward
- late entry
- higher timeframe conflict
- no validation
- choppy market
- low liquidity
- futures liquidation risk

Final rule:

Only A+ and A can be executable candidates.

B is watchlist/manual review.

C and Avoid become NO TRADE.

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PHASE 10 — LOW CONFIDENCE PROTECTION AUDIT

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Audit whether the system incorrectly shows BUY/SELL when confidence is very low.

Suggested confidence logic:

- confidence below 40% = NO TRADE
- confidence 40–60% = watchlist only

- confidence 60–75% = conditional setup
- confidence 75%+ = stronger setup, still risk filtered

Flag HIGH RISK if:

- system outputs BUY/SELL with confidence below 40%
- system shows 6%, 7%, 11% confidence but still says BUY
- confidence is not connected to historical win rate
- confidence is not connected to validation status
- confidence is not connected to risk/reward
- confidence is not connected to market regime

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PHASE 11 — SIGNAL BASKET / PORTFOLIO EXPECTANCY AUDIT

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When multiple signals are generated together, audit them as a basket.

Calculate:

- average potential gain to TP1
- average potential gain to TP2
- average potential loss to stop
- average risk/reward
- number of BUY/LONG signals
- number of SELL/SHORT signals
- asset concentration
- crypto exposure
- stock exposure
- futures exposure
- leverage exposure
- correlated asset exposure
- duplicate same-direction risk
- BTC/Nasdaq influence if crypto and tech stocks are mixed

Calculate expected basket return:

Expected Basket Return =

$(\text{win rate} \times \text{average TP gain}) - (\text{loss rate} \times \text{average stop loss})$

Run scenarios:

- 40% win rate
- 50% win rate
- 55% win rate
- 60% win rate
- 65% win rate

- 70% win rate

Output:

- breakeven win rate
- realistic expected return
- worst-case loss if all stops hit
- best-case gain if all TP1 hits
- whether basket is worth taking
- whether too many correlated trades exist

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PHASE 12 — PROFITABILITY REALISM AUDIT

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Do not optimize for fake 95% win rate.

Audit whether the system focuses on:

- positive expectancy
- low drawdown
- strong risk/reward
- low false-signal rate
- capital protection
- stable performance across regimes
- fewer but better trades

Check if the system tracks:

- win rate
- loss rate
- profit factor
- expectancy
- max drawdown
- Sharpe ratio
- Sortino ratio
- recovery factor
- average risk/reward
- average win
- average loss
- false signal rate
- signal frequency
- average hold time
- performance by symbol
- performance by timeframe
- performance by market regime

Flag HIGH RISK if:

- accuracy is claimed without proof
- 95%+ accuracy is displayed without evidence
- confidence and accuracy are treated as same
- win rate is optimized while drawdown is ignored
- risk/reward is ignored
- no slippage/spread/fees
- no walk-forward validation
- no out-of-sample validation

Important:

A 55% win rate with 2:1 risk/reward can be profitable.

A 90% win rate can still lose money if losses are large.

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PHASE 13 — BACKTEST / WALK-FORWARD / OUT-OF-SAMPLE AUDIT

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Audit whether the system proves its signals have edge.

Check:

- historical backtesting
- realistic entry simulation
- realistic exit simulation
- stop loss simulation
- TP1/TP2 simulation
- trailing stop simulation
- fees
- spread
- slippage
- partial fill assumptions
- candle-by-candle simulation
- lookahead bias prevention
- survivorship bias warning
- train/test split
- out-of-sample validation
- walk-forward validation
- paper trading logs
- live signal journal

For every strategy if possible report:

- win rate
- loss rate

- average win
- average loss
- average risk/reward
- expectancy
- profit factor
- max drawdown
- Sharpe
- Sortino
- recovery factor
- total trades
- false signal rate
- sample size
- market condition dependency

Flag HIGH RISK if:

- no backtesting
- no out-of-sample testing
- no paper logs
- no slippage/spread
- no commission model
- no drawdown reporting
- no expectancy calculation
- no sample-size requirement
- no lookahead-bias prevention

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PHASE 14 — SELF-LEARNING / FAILED SIGNAL FEEDBACK AUDIT

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I want the system to learn from failed signals safely.

Audit whether every signal is journaled with:

- signal ID
- symbol
- asset type
- trading mode
- timeframe
- signal
- entry price
- current price at signal
- stop loss
- take profit 1
- take profit 2
- invalidation level

- confidence
- bullishScore
- bearishScore
- conflictScore
- candle patterns detected
- indicator values
- strategy results
- market regime
- higher timeframe bias
- volume condition
- volatility condition
- macro/sentiment condition
- timestamp
- final trade decision
- setup grade

Audit whether outcome tracking records:

- TP hit?
- stop hit?
- invalidated?
- expired?
- max favorable excursion
- max adverse excursion
- time to outcome
- final profit/loss %
- failure reason

Audit whether failed signals are analyzed by:

- symbol
- timeframe
- strategy
- indicator combination
- candle pattern
- market regime
- volume condition
- volatility condition
- higher timeframe conflict
- spot vs futures mode
- leverage level
- session/news condition

Audit whether the system updates:

- strategy weights
- indicator weights

- pattern reliability
- confidence calibration
- symbol-specific reliability
- timeframe-specific reliability
- market-regime reliability
- no-trade thresholds
- false signal filters

Critical safety rules:

- do not update from one failed signal
- use rolling statistics
- require minimum sample size
- separate training and validation data
- use holdout validation
- use walk-forward validation
- keep model versions
- compare new model vs old model
- reject new model if worse
- never increase confidence without validation improvement

Required self-learning components:

- SignalJournal
- OutcomeTracker
- FeatureSnapshotStore
- StrategyPerformanceTracker
- IndicatorPerformanceTracker
- PatternPerformanceTracker
- ConfidenceCalibrator
- WeightUpdater
- ModelVersionRegistry
- WalkForwardValidator
- SafetyGate

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PHASE 15 — MARKET SENTIMENT / MACRO CORRELATION AUDIT

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Audit whether system checks broad market context.

For stocks:

- SPY direction
- QQQ/Nasdaq direction
- VIX level
- sector ETF direction if available

- earnings event risk
- Fed/macroeconomic event windows
- premarket gap risk
- news shock risk

For crypto:

- BTC trend
- ETH trend
- BTC dominance
- broad crypto market direction
- funding spikes if available
- volatility spikes
- liquidation cascade risk
- risk-on/risk-off macro sentiment

For both:

- broad market panic
- correlated asset collapse
- macro event window
- low liquidity period
- sudden volatility expansion

Required behavior:

- technical BUY can be blocked by macro danger
- weak altcoin BUY blocked if BTC is dumping
- tech stock BUY reduced if QQQ/SPY bearish
- futures confidence reduced during volatility shock

If live macro data is unavailable, module should be optional/configurable.

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PHASE 16 — RISK / CAPITAL PROTECTION AUDIT

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Audit whether every executable signal includes:

- entry zone
- stop loss
- take profit 1
- take profit 2
- invalidation level
- ATR-based stop
- structure-based stop
- risk/reward ratio
- trailing stop suggestion

- max loss %
- volatility warning
- liquidity warning
- spread warning
- slippage warning
- commission impact
- position size suggestion
- maximum risk per trade
- duplicate exposure warning
- correlated asset warning

Strict rules:

- no BUY/LONG without stop loss
- no SELL/SHORT without invalidation
- no trade if risk/reward is poor
- no trade if volatility is abnormal
- no trade if liquidity is weak
- no trade if spread is too high
- no trade if data is stale
- no trade if signal is unvalidated and confidence is low
- no futures trade if liquidation risk is dangerous

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PHASE 17 — DATA QUALITY / PROVIDER SAFETY AUDIT

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Audit candle data validation.

Check:

- missing candles
- duplicate candles
- stale candles
- bad timestamps
- future timestamps
- invalid OHLC
- zero volume
- negative prices
- outlier candles
- gaps in sequence
- timezone mismatch
- stock split/dividend adjustment
- crypto 24/7 continuity
- stock market session gaps
- provider retry handling

- provider failure handling
- API rate-limit handling
- stale cache handling

Bad data must force:
NO TRADE

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PHASE 18 — MARKET SESSION AUDIT

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For stocks:

- premarket
- regular session
- after-hours
- holidays
- half days
- market closed
- earnings windows
- news windows

For crypto:

- 24/7 trading
- weekend volatility
- exchange maintenance
- low liquidity hours
- funding-rate events if futures
- liquidation cascade risk

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PHASE 19 — COST MODEL AUDIT

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Check whether signal quality includes real trading costs:

- spread
- slippage
- commission
- partial fills
- liquidity risk
- funding fees for futures
- borrow/margin cost if applicable

A signal should be rejected if expected move is too small after costs.

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PHASE 20 — SPOT VS FUTURES MODE AUDIT

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Audit whether system separately handles SPOT and FUTURES.

SPOT mode:

- SPOT_BUY = long/accumulate
- SPOT_SELL_EXIT = exit/reduce spot
- SPOT_NO_TRADE
- no liquidation risk
- no funding fee
- no leverage
- no short unless margin explicitly supported
- risk mainly price drop and opportunity cost

FUTURES mode:

- FUTURES_LONG
- FUTURES_SHORT
- FUTURES_NO_TRADE
- leverage required
- isolated/cross margin support
- estimated liquidation price
- liquidation buffer
- funding fee risk
- maintenance margin risk
- forced liquidation protection
- max loss before stop
- stop distance vs liquidation distance

Strict futures rules:

- no futures signal without stop loss
- no futures signal without leverage input
- no futures signal without liquidation warning
- reject if stop loss is too close to liquidation
- reject if liquidation buffer is too small
- reject if volatility is extreme
- reduce confidence if funding/spread/slippage is unfavorable
- futures signals require stricter confirmation than spot

Required futures output:

- leverage
- marginType

- estimatedLiquidationPrice
- liquidationBufferPercent
- liquidationRisk
- fundingFeeRisk
- maxLossBeforeStop

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PHASE 21 — DEAD CODE / FAKE LOGIC / HARDCODE AUDIT

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Find and report:

- unused strategies
- disconnected indicators
- disconnected patterns
- imported but unused modules
- fake confidence
- hardcoded confidence
- random scores
- hardcoded stop loss
- hardcoded take profit
- fake probability
- placeholder ML model
- UI-only features
- duplicated logic
- unreachable code
- mocked-only tests
- tests that do not validate real behavior

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PHASE 22 — PROFESSIONAL SCORECARD

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Give scores out of 100:

- Candle Pattern Depth
- Indicator Depth
- Strategy Intelligence
- Bull/Bear Bias Quality
- Market Structure Quality
- Market Regime Intelligence
- Multi-Timeframe Quality
- Signal Selectivity
- Confidence Reliability
- Risk Protection

- Data Safety
- Cost Model Realism
- Spot/Futures Separation
- Futures Liquidation Safety
- Live Signal Executability
- Basket/Portfolio Risk Awareness
- Macro Sentiment Safety
- Backtesting Maturity
- Walk-Forward Validation
- Self-Learning Architecture
- Production Reliability
- Overall Profit Potential

Be brutally honest.

=====
STAGE 1 FINAL AUDIT REPORT FORMAT
=====

Return this report first:

INSTITUTIONAL TRADING SIGNAL ENGINE FORENSIC AUDIT

1. Executive Summary

Classify the system as:

- prototype
- hobby bot
- basic signal tool
- intermediate signal engine
- advanced selective signal engine
- near institutional-grade engine

2. Overall Honest Score /100

3. Full Feature Inventory

4. Signal-Only Safety Report

5. Bull/Bear Bias Truth Report

6. Candlestick Pattern Audit

7. Indicator Audit

- ## 8. Strategy Audit
 - ## 9. Market Structure Audit
 - ## 10. Market Regime Audit
 - ## 11. Multi-Timeframe Audit
 - ## 12. Final Signal Engine Truth Report
 - ## 13. Live Signal Executability Report
 - ## 14. Low Confidence Protection Report
 - ## 15. Basket / Portfolio Expectancy Report
 - ## 16. Profitability Realism Report
 - ## 17. Backtesting / Walk-Forward / Out-of-Sample Report
 - ## 18. Self-Learning / Failed Signal Feedback Report
 - ## 19. Macro Sentiment / Correlation Report
 - ## 20. Risk / Capital Protection Report
 - ## 21. Data Quality / Provider Safety Report
 - ## 22. Spot vs Futures Safety Report
 - ## 23. Dead / Fake / Hardcoded Code Report
 - ## 24. Exact Reasons This Engine May Still Lose Money
 - ## 25. Critical Missing Institutional Features
 - ## 26. Recommended Phase-Wise Upgrade Plan
 - ## 27. Final Recommendation
- Tell me honestly:
- keep current architecture
 - partial refactor
 - rebuild signal engine only

- rebuild strategy layer
- rebuild data layer
- add learning layer
- full rebuild only if necessary

After this audit, stop.
Do not implement yet.
Wait for my instruction.

=====

STAGE 2 — IMPLEMENTATION PLAN FORMAT

=====

When I say "CREATE IMPLEMENTATION PLAN", create a detailed implementation plan.

The plan must show:

- files to add
- files to modify
- existing modules to preserve
- compatibility adapters needed
- test strategy
- risk of breaking current system
- phase order
- expected output after each phase

Do not implement until I say:
START IMPLEMENTATION

=====

STAGE 3 — PHASE-WISE IMPLEMENTATION

=====

When I say "START IMPLEMENTATION", implement in this order.

Do not skip phases.
Do not jump randomly.
Do not rewrite entire system unless absolutely required.
Preserve current working behavior.
Every phase must include tests.

=====

IMPLEMENTATION PHASE 0 — BASELINE SAFETY

=====

Before coding:

- inspect current architecture briefly again
- identify current entry points
- identify current final signal engine
- identify current types
- identify current provider layer
- identify current tests
- create a safe branch/implementation checklist if git is available
- do not delete existing code
- do not remove current signal output

=====

IMPLEMENTATION PHASE 1 — CORE TYPES AND STANDARD OUTPUT CONTRACT

=====

Create or update shared types.

Candle:

- timestamp
- open
- high
- low
- close
- volume
- symbol
- timeframe
- source

Signal:

- symbol
- assetType: STOCK | CRYPTO
- tradingMode: SPOT | FUTURES
- timeframe
- signal:
 - SPOT_BUY
 - SPOT_SELL_EXIT
 - SPOT_NO_TRADE
 - FUTURES_LONG
 - FUTURES_SHORT
 - FUTURES_NO_TRADE
- direction: LONG | SHORT | EXIT | NO_TRADE
- confidence
- bullishScore

- bearishScore
- neutralScore
- conflictScore
- upsideProbability
- downsideProbability
- riskLevel
- signalQuality
- validationStatus
- setupGrade
- tradeDecision
- entryZone
- stopLoss
- takeProfit1
- takeProfit2
- riskRewardRatio
- invalidation
- reasoning
- failedConfirmations
- marketRegime
- higherTimeframeBias
- dataQualityStatus
- timestamp

StrategyResult:

- strategyName
- signalCandidate
- confidence
- bullishScore
- bearishScore
- reasons
- failedConfirmations
- requiredConfirmations
- riskNotes
- validationStatus

PatternResult:

- patternName
- bias
- category
- strength
- reliability
- explanation
- candleIndex
- timestamp

IndicatorResult:

- name
- value
- bias
- strength
- explanation

RiskResult:

- entry
- stopLoss
- takeProfit1
- takeProfit2
- riskRewardRatio
- invalidationLevel
- maxLossPercent
- volatilityWarning
- liquidityWarning
- spreadWarning
- slippageWarning

FuturesRiskResult:

- leverage
- marginType
- estimatedLiquidationPrice
- liquidationBufferPercent
- liquidationRisk
- fundingFeeRisk
- maxLossBeforeStop

Add tests for output shape and compatibility.

=====

IMPLEMENTATION PHASE 2 — CANDLE DATA VALIDATION ENGINE

=====

Implement CandleValidator.

It must check:

- missing candles
- duplicate timestamps
- stale candles
- future timestamps
- invalid OHLC

- high lower than open/close
- low higher than open/close
- zero volume
- negative prices
- outlier candles
- gaps in sequence
- timezone mismatch
- stock session gaps
- crypto continuity issues

Output:

- isValid
- severity
- warnings
- errors
- shouldBlockSignal

Rule:

Critical data issues force final NO_TRADE.

Add tests for every validation case.

=====

IMPLEMENTATION PHASE 3 — FULL CANDLESTICK PATTERN ENGINE

=====

Implement full candlestick pattern detection.

Patterns:

- Doji
- Spinning Top
- Marubozu
- Hammer
- Inverted Hammer
- Shooting Star
- Hanging Man
- Bullish Engulfing
- Bearish Engulfing
- Inside Bar
- Outside Bar
- Morning Star
- Evening Star
- Three White Soldiers
- Three Black Crows

- Tweezer Top
- Tweezer Bottom
- Piercing Pattern
- Dark Cloud Cover
- Bullish Harami
- Bearish Harami
- Pin Bar
- Rejection Candle
- Breakout Candle
- Exhaustion Candle
- Momentum Candle
- Long Wick Candle
- Narrow Range Candle
- Wide Range Candle
- Trap Candle
- Failed Breakout Candle

Each pattern must return:

- name
- bullish/bearish/neutral bias
- reversal/continuation/indecision type
- strength 0–100
- reliability 0–100
- explanation
- candle index
- timestamp

Important:

Pattern alone must not create BUY/SELL.

It only contributes to confluence.

Add tests for each pattern using sample candles.

=====

IMPLEMENTATION PHASE 4 — INDICATOR ENGINE

=====

Implement or complete indicators.

Trend:

- SMA
- EMA
- WMA
- HMA

- Supertrend
- Ichimoku
- Parabolic SAR

Momentum:

- RSI
- MACD
- Stochastic RSI
- Stochastic Oscillator
- CCI
- ROC
- Momentum

Volatility:

- ATR
- Bollinger Bands
- Keltner Channels
- Donchian Channels
- Standard Deviation
- Bollinger Band Width

Volume:

- VWAP
- OBV
- Volume Moving Average
- Volume Spike
- MFI
- Accumulation/Distribution

Key levels:

- Pivot Points
- Fibonacci Retracement
- Fibonacci Extension
- Swing High/Low
- Previous High/Low
- Supply/Demand Zones

Each indicator must output:

- value
- bullish/bearish/neutral interpretation
- strength score
- explanation
- configurable period
- no hardcoded-only settings

Add formula tests for major indicators.

=====

IMPLEMENTATION PHASE 5 — PRICE ACTION / MARKET STRUCTURE ENGINE

=====

Implement:

- higher highs
- higher lows
- lower highs
- lower lows
- break of structure
- change of character
- support/resistance
- swing points
- retest detection
- breakout detection
- fake breakout detection
- liquidity sweep
- consolidation/range detection
- trendline rejection
- supply/demand zone reaction

Output:

- trendBias
- structureBias
- keyLevels
- invalidationLevel
- breakoutStatus
- liquidityStatus
- explanation

Add tests with synthetic candle sequences.

=====

IMPLEMENTATION PHASE 6 — BULLISH / BEARISH BIAS ENGINE

=====

Create BullBearBiasEngine.

It must calculate:

- candleBiasScore
- trendBiasScore

- momentumBiasScore
- volumeBiasScore
- higherTimeframeBiasScore
- structureBiasScore
- bullishScore
- bearishScore
- neutralScore
- conflictScore

Decision:

- bullish dominant = BUY/LONG candidate
- bearish dominant = SELL/SHORT candidate
- mixed = NO_TRADE

Rules:

- one indicator cannot trigger trade alone
- higher timeframe conflict reduces confidence
- high conflict forces NO_TRADE
- neutral/choppy condition forces NO_TRADE

Add tests:

- strong bullish setup
- strong bearish setup
- mixed setup
- HTF conflict setup
- choppy setup

=====

IMPLEMENTATION PHASE 7 — STRATEGY MODULES

=====

Implement strategy modules.

Core:

- EMA crossover
- SMA crossover
- RSI reversal
- MACD momentum
- Bollinger mean reversion
- VWAP reclaim/rejection
- support/resistance breakout
- pullback continuation
- candlestick reversal
- trend following

- volume breakout
- ATR breakout
- multi-timeframe alignment

Advanced:

- liquidity sweep reversal
- order block reaction
- fair value gap reaction
- Fibonacci retracement bounce
- opening range breakout
- gap fill
- break and retest
- range breakout
- range fade
- scalping
- swing continuation
- momentum continuation
- volatility contraction breakout
- failed breakout reversal

Each strategy returns:

- strategyName
- signal candidate
- bullishScore
- bearishScore
- confidence
- required confirmations
- passed confirmations
- failed confirmations
- risk notes
- invalidation idea
- validation status

Strategies produce candidates only.

Final engine decides executable signal.

Add tests for each strategy.

```
=====
IMPLEMENTATION PHASE 8 — MARKET REGIME ENGINE
=====
```

Implement MarketRegimeEngine.

Detect:

- bullish trending
- bearish trending
- sideways/range
- choppy/noise
- breakout regime
- mean reversion regime
- high volatility
- low volatility
- low liquidity
- news shock mode if data available

Use:

- ADX
- ATR
- Bollinger width
- volume
- trend slope
- range compression
- breakout expansion

Rules:

- trend strategies enabled in trend
- mean-reversion strategies enabled in range
- breakout strategies need volume expansion
- choppy/noisy regime forces NO_TRADE or reduces confidence

Add tests.

=====

IMPLEMENTATION PHASE 9 — MULTI-TIMEFRAME ENGINE

=====

Implement MultiTimeframeEngine.

Mapping:

- 1m checks 5m and 15m
- 5m checks 15m and 1h
- 15m checks 1h and 4h
- 1h checks 4h and 1d
- 4h checks 1d and 1w

Output:

- lower timeframe bias
- higher timeframe bias
- alignment score
- conflict score
- recommendation

Rules:

- HTF alignment increases confidence
- HTF conflict reduces confidence
- strong HTF conflict blocks weak trades

Add tests.

```
=====
IMPLEMENTATION PHASE 10 — CONFLUENCE ENGINE
=====
```

Create ConfluenceEngine.

It combines:

- candle patterns
- indicators
- strategies
- price action
- market regime
- higher timeframe
- volume
- risk/reward
- data quality

Output:

- final bullish score
- final bearish score
- neutral/conflict score
- confidence
- reasoning
- failed confirmations

Rules:

- no trade from single confirmation
- no trade if conflict too high
- no trade if RR poor
- no trade if data invalid
- no trade if regime does not support strategy

- no trade if confidence below threshold

Thresholds:

- below 40 = NO_TRADE
- 40–60 = watchlist
- 60–75 = conditional
- 75+ = stronger, still risk filtered

=====

IMPLEMENTATION PHASE 11 — RISK ENGINE

=====

Implement RiskEngine.

Calculate:

- entry zone
- ATR-based stop
- structure-based stop
- invalidation level
- TP1
- TP2
- risk/reward ratio
- trailing stop suggestion
- max loss %
- volatility warning
- liquidity warning
- spread/slippage warning
- commission impact

Rules:

- no BUY/LONG without stop loss
- no SELL/SHORT without invalidation
- reject if RR below configured minimum
- reject if stop is unrealistic
- reject if expected move is too small after costs

Add tests.

=====

IMPLEMENTATION PHASE 12 — SPOT VS FUTURES RISK ENGINE

=====

Implement separate logic for SPOT and FUTURES.

SPOT:

- SPOT_BUY = long/accumulate
- SPOT_SELL_EXIT = exit/reduce
- SPOT_NO_TRADE
- no liquidation risk
- no funding fee
- no leverage

FUTURES:

- FUTURES_LONG
- FUTURES_SHORT
- FUTURES_NO_TRADE
- requires leverage input
- supports isolated/cross margin
- estimates liquidation price
- calculates liquidation buffer
- checks funding fee risk
- checks margin risk
- rejects if liquidation risk is high
- rejects if stop loss is too close to liquidation

Futures output:

- leverage
- marginType
- estimatedLiquidationPrice
- liquidationBufferPercent
- liquidationRisk
- fundingFeeRisk
- maxLossBeforeStop

Rule:

Futures safety must be stricter than spot.

Add tests:

- spot buy
- spot exit
- futures long
- futures short
- liquidation danger rejection

=====
IMPLEMENTATION PHASE 13 — LIVE SIGNAL EXECUTABILITY FILTER
=====

Implement TradeDecisionLayer.

Final decisions:

- TAKE
- WAIT
- SKIP
- NO_TRADE

For every signal calculate:

- entry-to-current distance
- TP1 %
- TP2 %
- stop %
- RR
- late entry status
- overextended status
- current validity

Rank setup:

- A+
- A
- B
- C
- Avoid

Rules:

- A+/A = executable candidate
- B = manual review/watchlist
- C/Avoid = NO_TRADE
- late entry = WAIT or SKIP
- low confidence = NO_TRADE
- weak validation = watchlist only

=====
IMPLEMENTATION PHASE 14 — PORTFOLIO / BASKET EXPECTANCY ENGINE
=====

Implement BasketRiskEngine.

When multiple signals appear, calculate:

- average TP1 %
- average TP2 %
- average stop %

- average RR
- worst-case if all stops hit
- best-case if all TP hits
- breakeven win rate
- expected return at 40%, 50%, 55%, 60%, 65%, 70% win rates
- asset concentration
- crypto exposure
- stock exposure
- futures exposure
- leverage exposure
- correlation risk
- duplicate direction risk

Warn if:

- too many crypto longs
- too many tech longs
- BTC/SPY/QQQ risk is high
- correlated assets are stacked
- same strategy fires on too many assets

=====

IMPLEMENTATION PHASE 15 — MACRO / SENTIMENT FILTER

=====

Implement optional MacroSentimentFilter.

For stocks:

- SPY trend
- QQQ trend
- VIX level
- earnings window
- Fed/macro calendar if available

For crypto:

- BTC trend
- ETH trend
- BTC dominance
- funding spikes if available
- broad crypto market risk

Rules:

- technical BUY can be blocked by broad market panic
- weak altcoin BUY blocked if BTC is dumping
- tech stock BUY reduced if QQQ/SPY bearish

- futures confidence reduced during volatility shock

If live macro data unavailable, make module optional/configurable.

=====

IMPLEMENTATION PHASE 16 — BACKTEST ENGINE

=====

Implement or upgrade BacktestEngine.

Must support:

- strategy backtest
- confluence backtest
- candle-by-candle simulation
- entry/exit simulation
- stop loss
- TP1/TP2
- trailing stop
- fees
- slippage
- spread
- partial fill assumption
- no lookahead bias

Metrics:

- win rate
- loss rate
- average win
- average loss
- average RR
- expectancy
- profit factor
- max drawdown
- Sharpe
- Sortino
- recovery factor
- total trades
- false signal rate
- performance by symbol
- performance by timeframe
- performance by regime

Add tests for lookahead-bias prevention.

=====

IMPLEMENTATION PHASE 17 — WALK-FORWARD / OUT-OF-SAMPLE VALIDATION

=====

Implement ValidationEngine.

Required:

- train/test split
- out-of-sample validation
- walk-forward windows
- strategy comparison
- overfitting warning
- minimum sample size requirement

Rules:

- strategy cannot be marked VALIDATED without enough trades
- unvalidated strategy cannot produce high confidence
- poor validation reduces or blocks signal

=====

IMPLEMENTATION PHASE 18 — SIGNAL JOURNAL / OUTCOME TRACKER

=====

Implement SignalJournal.

Store every generated signal with:

- symbol
- timeframe
- asset type
- trading mode
- signal
- confidence
- candle snapshot
- indicators
- patterns
- strategy results
- market regime
- HTF bias
- entry
- stop
- TP
- timestamp
- final decision

- setup grade
- outcome later

Implement OutcomeTracker:

- hit TP?
- hit stop?
- expired?
- invalidated?
- max favorable excursion
- max adverse excursion
- time to outcome
- failure reason

=====

IMPLEMENTATION PHASE 19 — SELF-LEARNING / ADAPTIVE WEIGHT SYSTEM

=====

Implement safe self-learning.

Create:

- StrategyPerformanceTracker
- IndicatorPerformanceTracker
- PatternPerformanceTracker
- ConfidenceCalibrator
- WeightUpdater
- ModelVersionRegistry
- SafetyGate

Learning rules:

- do not update from one failed signal
- use rolling windows
- require minimum sample size
- update symbol-specific stats
- update timeframe-specific stats
- update market-regime-specific stats
- compare new weights vs old weights
- reject new model if worse
- keep model versions
- never increase confidence without validation

The system should learn:

- which strategies fail often
- which indicators add value
- which candle patterns are unreliable

- which market regimes are dangerous
- which symbols/timeframes perform best

=====

IMPLEMENTATION PHASE 20 — CONFIDENCE CALIBRATION ENGINE

=====

Implement confidence calibration.

Confidence should be based on:

- confluence strength
- historical strategy reliability
- symbol reliability
- timeframe reliability
- market regime reliability
- risk/reward
- HTF alignment
- recent false signal rate
- validation status

Rules:

- unvalidated = confidence cap
- high conflict = confidence reduction
- low RR = confidence reduction
- bad regime = confidence reduction
- good backtest + good paper logs = confidence boost

No arbitrary confidence.

=====

IMPLEMENTATION PHASE 21 — API / CLI OUTPUT

=====

Expose signal output through current interface.

CLI example:

```
npm run analyze -- --symbol BTCUSDT --timeframe 15m --mode FUTURES --  
leverage 3 --margin isolated
```

Output:

- signal
- trade decision
- setup grade
- confidence

- bullish/bearish score
- risk/reward
- stop
- TP
- liquidation risk if futures
- data quality
- market regime
- reasoning
- failed confirmations

JSON output must match the standard contract.

=====

IMPLEMENTATION PHASE 22 — TESTING / QUALITY GATES

=====

Add tests for:

- candle validator
- candle patterns
- indicators
- strategies
- bull/bear bias
- market structure
- confluence
- risk engine
- futures risk
- market regime
- backtest
- validation
- self-learning
- confidence calibration
- final signal output

Quality gates:

- npm test
- typecheck
- lint if available
- no failing tests before moving phases

=====

IMPLEMENTATION PHASE 23 — FINAL INTEGRATION RULES

=====

Final engine must obey:

1. Bad data = NO_TRADE
2. Low confidence = NO_TRADE or watchlist
3. Poor RR = NO_TRADE
4. HTF conflict = reduce/block
5. Choppy regime = reduce/block
6. Futures risk high = FUTURES_NO_TRADE
7. Unvalidated strategy = confidence cap
8. Single indicator cannot trigger trade
9. Weak signal cannot be executable
10. Every signal must have reasoning
11. Every signal must be journaled
12. Every outcome must improve future statistics safely
13. Spot and futures must use separate risk logic
14. Macro danger can block weak technical signals
15. Self-learning cannot update without enough samples

=====

IMPLEMENTATION OUTPUT FORMAT

=====

For each implementation phase, output:

Phase Name

- Files added
- Files modified
- What was implemented
- Tests added
- How to run tests
- Any known limitations
- Next phase recommendation

Stop after each major phase if changes are large.

Summarize progress before continuing.

Do not break existing signal output.